

轮 趣 科 技

WHEELTEC Self-Balancing Bicycle Hands-on Use

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Imprint:

releases	dates	Description of content
V1.0	2022/06/23	First Release

catalogs

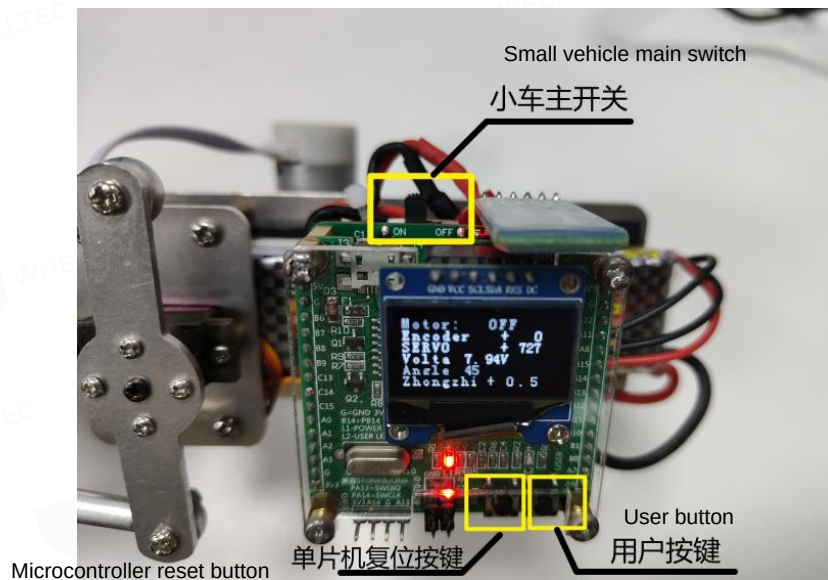
1. Getting Started.....	3
1.1 Trolley Balance Function Use.....	3
1.2 Trolley Display Parameters.....	3
1.3 Bluetooth Remote Control Car	4
1.4 Viewing Waveforms with a PC	5
1.5 Trolley balance effect adjustment	6
1.6 Trolley Charging.....	7
1.7 Download program.....	7

1. get started

1.1 Trolley balance function use

Upon receipt of the bike, make sure that there is no damage to the exterior, then you can start using it.

First of all, turn on the main switch of the trolley, at this time, the indicator light and display screen above the trolley will be lit. There are two ways to start the trolley: one way is to hold the trolley upright on the ground and press the user button to start the trolley; the other way is to



put the trolley on the ground and push it forward to start the trolley. When the trolley is tilted to the left or right by more than 40° , the rear wheel motor will turn off automatically. The main switch and user button are shown in Figure 1-1.

Figure 1-1 Schematic diagram of the main board of the trolley

1.2 Trolley Display Parameters

The OLED display data for the cart is described below:



- 第一行显示电机的状态
- 第二行显示后轮编码器的读数
- 第三行显示小车舵机输出的PWM数值
- 第四行显示小车的电池电压
- 第五行显示小车左右倾斜的角度
- 第六行显示小车的平衡中值

The first line displays the status of the motor.
The second line shows the readings from the rear wheel encoder.
The third line displays the PWM values output by the cart's servo.
The fourth line shows the battery voltage of the cart.
The fifth line displays the left and right tilt angles of the cart.
The sixth line shows the balance midpoint of the cart.

Figure 1-2 Cart OLED Display Value Description

1.3 Bluetooth remote control car

① Controlling the trolley

Install MiniBalance APP on your cell phone, search for BT04-A in the APP and connect it. After the connection is completed, you can see the data returned from the car in the Debug column, which means it has been successfully connected. Now put the cart on the ground and turn on the balance, you can control the movement of the cart in the APP.

The car only supports joystick to control the direction, can not control the acceleration and deceleration, as well as backward, of which the control description is as follows:



Figure 1-3 Mobile APP Interface

APP Rocker	↑	↓		←	↙		→	↘
Trolley status	(math.) invariant	(math.) invariant	leftwards	leftwards	leftwards	the right	the right	the right

Table 1-1 Mobile phone APP operation description

②Use APP to view waveforms and adjust parameters online.

Click "Waveform" to enter the waveform view interface, which has three

parameters for users to view, namely the angle of the car, angular velocity and the PWM value of the input servo.

Click "Debug" to enter the adjustment interface, before debugging, you need to click on the three dots in the upper right corner of the page.

Then click Get Device Parameters to get the data of the trolley, in which the parameters of the debugging interface are described as follows:

The first three rows are the actual magnitudes of K_p , K_i , and K_d adjusted for the LQR error, respectively; the fourth row is the trolley The mechanical median value of the body inclination angle after enlarging 100 times, the value can be adjusted by the potentiometer on the main board; rows 5 and 6 are the left and right steering amplitude after



enlarging 10 times, when the user adjusts the value, the input value will be scaled down by 10 times and then assigned again in the program.

Figure 1-4 Mobile APP Online Parameter Adjustment

1.4 Viewing Waveforms with a PC

By default, the trolley uses APP to send data, if you need to use PC to view the waveform, you need to double-click the user button on the trolley's main board, and then you will see the OLED screen stops refreshing, which means the trolley enters the PC waveform display mode.

Before using the PC to view the waveform, you need to connect the

trolley to the computer with a data cable, the connection instructions are shown in Figure 1-5. We find MiniBalance waveform display host software in the catalog "6.Software and Driver/6.MiniBalance Host Software", right click to open it as administrator, select the corresponding serial port, choose 128000 baud rate, and click "Open Serial Port" to see the waveform. serial port", **then** you can see the waveform. By default, the waveform displays the left and right tilt angle of the vehicle, the angular velocity of the left and right tilt angles, and the PWM value input to the servo, as shown in Figure 1-6.

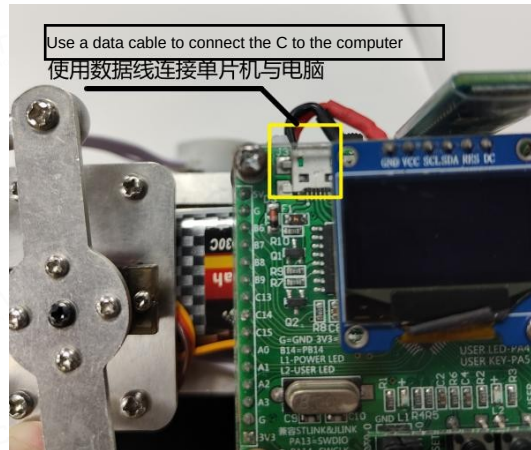


Figure 1-5 Connection Prompt Diagram

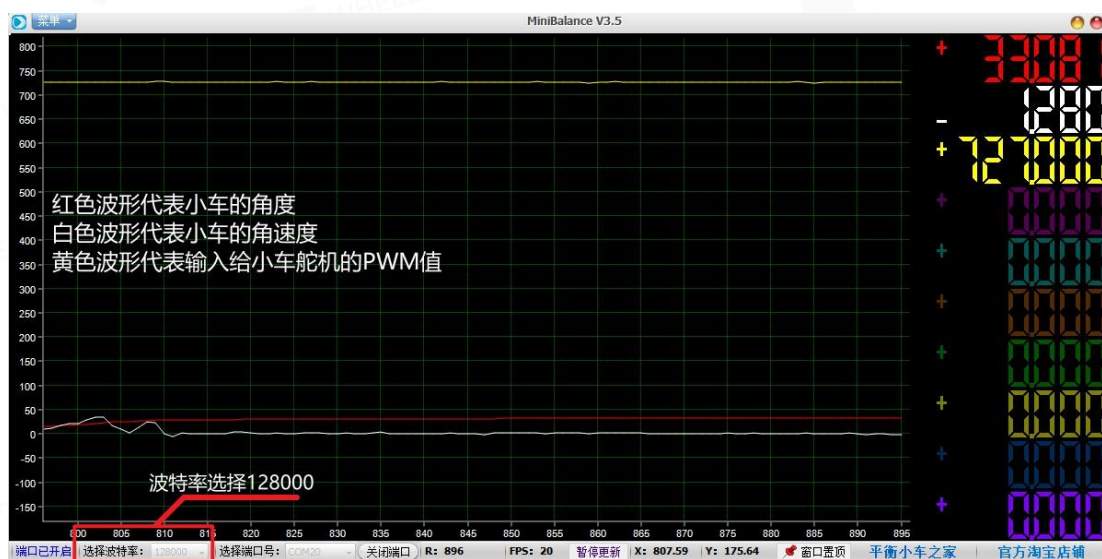


Figure 1-6 Waveform Display on PC

1.5 Trolley balance effect adjustment

If the trolley is not well balanced and will fall down during operation, the effect can be optimized by adjusting the two potentiometers on the main board of the trolley. Note: We will debug the trolley at the factory before shipment, users generally do not need to adjust themselves.

Faucet Adjustment: Twist the potentiometer on the left side and adjust it until the faucet is facing straight ahead.

Adjustment of balance center value: Place the trolley in a balanced upright position, observe the value of "Angle" on the display, when the trolley is placed vertically, this value is the balance center value, twist the potentiometer on the right side to adjust to this value. Adjustment example: if the cart is placed vertically balanced, "Angle" shows the value of 1, then twist the potentiometer to adjust the value of "Zhongzhi" to 1; if the cart is

placed vertically balanced, "Angle" shows the value of 1. If the value of "Angle" is 359 when the trolley is balanced vertically, then turn the potentiometer to adjust the value of "Zhongzhi" to -1.

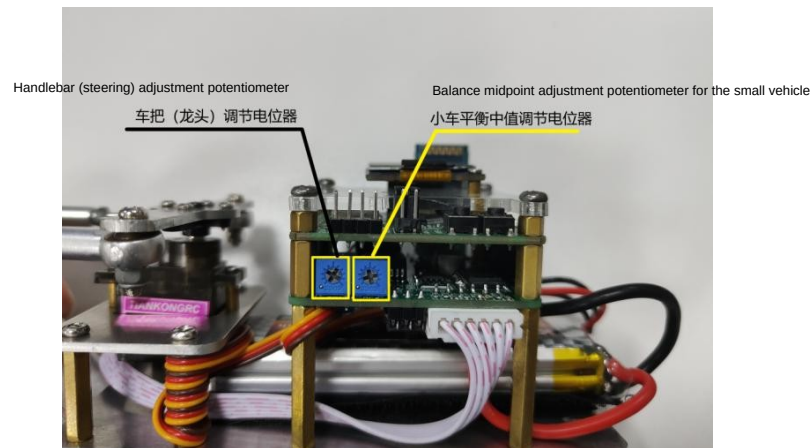
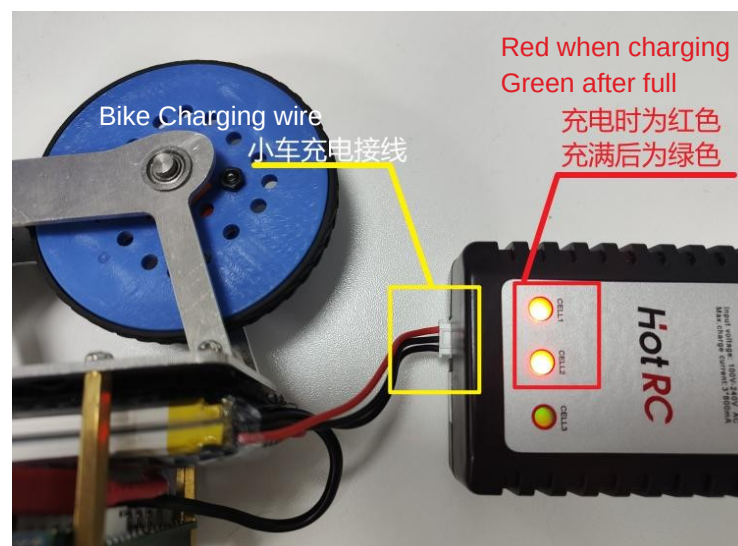


Figure 1-7 Trolley Potentiometer Position Diagram

1.6 Trolley charging

The trolley is powered by a 2s modeling battery. If the battery is over-discharged, it will cause permanent damage to the battery. The trolley is equipped with a voltage detection function, which will automatically turn off the trolley's motor when the voltage shown on the display is lower than 7.1V. When the voltage on the display falls below 7.1V, the cart's motor will automatically shut off. When the voltage is below 7.1V, the cart should be recharged in a timely manner.

When charging, the two indicator lights on the charger are red and green when full. Note: It is normal for the charger to get hot when charging with high current.





1.7 download

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The motherboard uses a one-click download circuit, which makes it very easy to download the program. All you need is a MicroUSB cell phone cable.

① Hardware Preparation

1. Trolley motherboard
2. MicroUSB cell phone data cable

② Software Preparation

1. FlyMCU burning software (included information there)
2. the corresponding USB to TTL module CH340G driver (the attached information also has a driver to provide) driver



installation is successful, you can open the device manager to view the

③ Wiring

It's very simple, the cable connects the computer to the board.

④ FlyMcu software setup, open the FlyMcu software in the attached file and make the following settings.

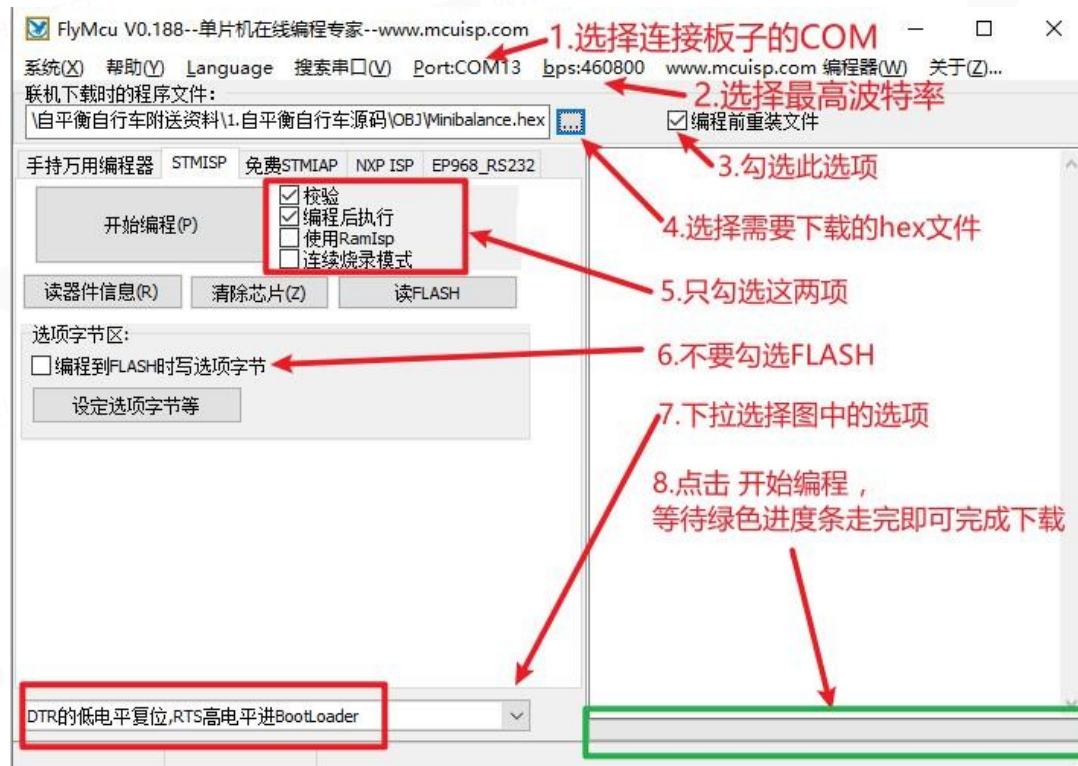


Figure 1-9 FlyMcu Software Setup Details

OK, everything is ready, then click Start Programming and the program will download! Because the Execute after programming is checked, the program will run automatically after it is downloaded.

